

AGROSPEAR CROPCARE CO., LTD.

ACARICIDES & NEMATOCIDES QUALITY CONTROL SYSTEM & DOSSIER

杀螨剂 / 杀线虫剂制剂生产全流程质量控制体系与佐证报告规程

Document Code: SOP-QC-ACN-2026

Effective Date: August 2026

Version: Rev 4.0 (Global Standard)

Compliance: FAO/WHO Specs & CIPAC Methods

Product Lines: SC, EC, GR, CS, EW Specialized Formulations

ACARICIDE / NEMATOCIDE SPECIALIZED SAFETY PROTOCOL: Nematicides (e.g., organophosphates/avermectins) involve higher soil persistence, root-zone penetrability, and occupational toxicity requirements. Granular nematicides (GR) must strictly ensure zero free dust (<0.5%) to prevent inhalation toxicity. Acaricide SCs (e.g., Spirodiclofen, Bifenazate) require precise crystal habit monitoring to prevent Ostwald ripening and quick egg-killing efficacy loss.

杀螨剂 / 杀线虫剂专业品控与安全特别规程：杀线虫剂（含有机磷类、阿维菌素类、噻唑膦等）对根系下渗传导性、土壤吸附性及职业防护具有极高安全要求；颗粒剂（GR）必须严格控制游离粉尘（<0.5%）以杜绝吸入中毒风险。杀螨剂悬浮剂（螺螨酯、联苯腈酯等）须严密监控晶型转化与奥氏熟化，确保卵-螨全时期速效与持效性。

1. END-TO-END QA/QC WORKFLOW / 杀螨剂与杀线虫剂生产品控全生命周期流程

QC Stage / 阶段	Key Control Parameters & Protocols / 关键管控项目与规范	Methodology / 检验方法与仪器	Release Criteria / 准则
IQC (Incoming QC) 原辅料进厂检验	• Technical Material (TC): AI content, stereoisomers (e.g., Spirotetramat / Pyridaben cis-trans purity), toxic impurities, free acidity. • 原药(TC): 有效成分纯度、顺反立体异构体比例（如螺虫乙酯/哒螨灵异构体）、微量酸度与水分。 • Adjuvants & Penetrants: Root-zone penetrants (silicone/fatty alcohol ethoxylates), vegetable oil carriers, clay/bentonite carrier hardness & bulk density (for GR). • 助剂与载体: 根际渗透剂、植物油增效剂、膨润土/凹凸棒土载体颗粒强度与堆积密度（GR 颗粒剂）。 • Barrier Packaging: UN-certified anti-puncture multi-layer bags (aluminum-plastic) for GR, fluorinated HDPE bottles for EC/SC. • 专用包材: UN 认证抗穿刺铝塑复合防潮袋（用于颗粒剂）、耐腐蚀氟化 HDPE 瓶（用于悬浮剂/乳油）。	• Chiral HPLC / GC-FID • Granule Hardness Tester • Karl Fischer Titrator • UN Drop/Leakage Test	100% Passed 不合格严禁入库
IPQC (In-Process) 制程过程巡检	• Acaricide SC Milling: Chill jacket outlet temp < 32°C (preventing heat degradation of Bifenazate), target fineness D90 < 3.0µm, crystal seed stabilization. • 杀螨悬浮剂研磨: 研磨出料温度 < 32°C（防止联苯腈酯等热敏原药分解），微粒细度 D90 < 3.0µm，晶型防结块稳定控制。 • Nematicide Granule (GR): Rotary drum impregnating & coating, binder uniformity, fluid bed low-temp curing (< 45°C), de-dusting efficiency. • 杀线虫颗粒剂包衣: 滚筒吸附/包膜喷雾均匀度、黏合剂固化干燥（床温 < 45°C）、负压多级除尘。 • Microencapsulated CS (Nematicide): Sustained-release wall thickness, unencapsulated AI < 1.5%. • 微囊悬浮剂(CS): 控释囊壁厚度测定、游离未包封原药残留 < 1.5%。 • Packaging Line: Seal air-tightness, net weight precision (±0.5%), 2D QR serialization. • 灌装与包装线: 铝塑袋/瓶口密封性负压检测、装量精密度抽检与防伪喷码。	• Laser Particle Sizer • CIPAC MT 171 (Dustmeter) • In-line Checkweigher • Brookfield Viscometer	CCP In-Spec 参数异常停机纠偏
FQC / OQC 成品终检与放行	• Multi-active Assay: HPLC-DAD baseline separation of dual/triple combinations (e.g., Bifenazate + Etoxazole / Fosthiazate + Abamectin). • 复配有效成分分析: 杀螨剂（联苯腈酯+乙螨唑）/ 杀线虫剂（噻唑膦+阿维菌素）色谱完全分离，符合 FAO 允差。 • Physical Properties: Suspensibility ≥ 93.0%, Granule attrition resistance ≥ 98.0%, spontaneous dispersion, Persistent foam ≤ 25 mL. • 理化性能检测: 悬浮率 ≥ 93.0%、颗粒耐磨碎率 ≥ 98.0%、水中自发崩解、持久起泡性 ≤ 25 mL。 • Stability Stress: Accelerated thermal stability (54±2°C, 14 days) & cold storage (0±2°C, 7 days) without active degradation (>3%) or caking. • 极限破坏试验: 54°C 热储 14 天原药降解率 ≤ 3.0%，0°C 冷储 7 天无结晶、无团聚。	• CIPAC MT 184 / MT 178.2 • CIPAC MT 46.3 / MT 39.3 • Reverse-Phase HPLC	COA Released QA 总监签字放行
Traceability & Safety 留样与职业防护追溯	• Toxicity containment verification: Plant ambient air organophosphate vapor monitoring (<0.01 mg/m³). • 职业暴露监测: 杀线虫剂车间环境空气有机磷/生物碱挥发度检测（<0.01 mg/m³）。 • Retained samples archived at controlled 20±2°C for shelf-life + 1 year (min. 3 years) linked to digital ERP batch dossiers. • 恒温留样库保存至保质期后 1 年，与 ERP 数字化批生产记录（BMR）及检验图谱绑定追溯。	• VOC Air Detector • Cloud ERP Traceability	Archived 建立终身追溯档案

2. FORMULATION-SPECIFIC QC STANDARDS (FAO/CIPAC COMPLIANT) / 主要杀螨剂与杀线虫剂剂型质控指标

Formulation / 剂型	Representative Products / 代表品种	Critical Quality Attributes (CQA) / 核心理化质控指标	Key Test Method / 对应方法
SC (Suspension Conc.) 杀螨悬浮剂	- Bifenazate 240g/L + Etoxazole 110g/L SC - Spirodiclofen 240g/L SC - Cyenopyrafen 30% SC	- Active Content: Nominal $\pm$ 5.0% - Particle Size: $D_{50} \leq 1.5\mu\text{m}$, $D_{90} \leq 3.0\mu\text{m}$ - Suspensibility: $\geq 93.0\%$ (after 54°C thermal test) - Pourability Residue: $\leq 2.5\%$, Rinsed $\leq 0.3\%$	- CIPAC MT 184 - Laser Diffraction - CIPAC MT 148.1
GR (Granules) 杀线虫颗粒剂	- Fosthiazate 10% GR - Fosthiazate 5% + Abamectin 0.5% GR - Fluopyram 2% GR	- Free Dust Content: $\leq 0.20\%$ (CIPAC MT 171) - Attrition Resistance: $\geq 98.0\%$ (CIPAC MT 178.2) - Slow-release in Soil: Consistent 30-day profile - Granule Size Range: 0.5 – 1.6 mm $\geq 95.0\%$	- CIPAC MT 171 - CIPAC MT 178.2 - Soil Leaching Model
EW / EC (Emulsion) 杀螨乳油 / 水乳剂	- Pyridaben 20% WP/EC - Propargite 570g/L EC - Fenbutatin Oxide 550g/L SC/EC	- Emulsion Stability: 0.5h/2h/24h no cream/oil (<1mL) - Cold Stability: 0$\pm$2°C for 7 days (transparent) - Moisture Content: $\leq 0.35\%$ (EC) - Flash Point: $\geq 65^\circ\text{C}$ (Safety storage)	- CIPAC MT 36.3 - CIPAC MT 39.3 - Tag Closed Cup
CS / ME (Controlled Release) 杀线虫微囊 / 微乳剂	- Abamectin 2% ME (Soil Drench) - Fluopyram 200g/L + Abamectin 50g/L SC/CS	- Encapsulation Efficiency: $\geq 98.0\%$ - Free Active Content: $\leq 1.5\%$ - Root Zone Mobility Index: Standard penetration	- Extraction HPLC - CIPAC MT 190

3. SUPPORTING EVIDENCE 1: OFFICIAL COA TEMPLATE / 佐证文件一：成品出厂检验报告单 (COA)

AGROSPEAR CROPCARE CO., LTD.

CERTIFICATE OF ANALYSIS / 产品检验分析报告单 (ACARICIDE / NEMATOCIDE)

Product Name / 品名: Bifenazate 240g/L + Etoxazole 110g/L SC (联苯肼酯 240g/L + 乙螨唑 110g/L 悬浮剂)

Batch No. / 生产批号: AS-ACN-20260826-11

Invoice / Order No. / 订单号: AGR-EXP-2026-950

Batch Quantity / 批量: 8,000 Liters (1L x 12/Carton)

Manufacture Date / 生产日期: 2026-08-26

Expiry Date / 有效期至: 2028-08-25

Sampling Date / 取样日期: 2026-08-26

Inspection Date / 判定日期: 2026-08-27

Test Parameter / 检验项目	Specification Limit / 标准指标	Test Result / 实测结果	Standard Method / 检验方法	Verdict / 判定
Appearance / 外观性状	Off-white homogeneous flowable suspension (类白色均质悬浮液体)	Conforms / 符合要求	Visual Check	Pass
Active Ingredient 1: Bifenazate (联苯肼酯)	240.0 $\pm$ 12.0 g/L (228.0 – 252.0 g/L)	242.1 g/L	HPLC-DAD (Agilent 1260)	Pass
Active Ingredient 2: Etoxazole (乙螨唑)	110.0 $\pm$ 5.5 g/L (104.5 – 115.5 g/L)	111.4 g/L	HPLC-DAD (Agilent 1260)	Pass
pH Value (1% aqueous solution)	5.5 – 8.0	6.6	CIPAC MT 75.3	Pass
Suspensibility / 悬浮率	$\geq 92.0\%$ (CIPAC Standard Water D)	97.8 %	CIPAC MT 184	Pass
Particle Size Distribution / 粒度 (D90)	$\leq 3.0\mu\text{m}$	2.1 μm (D50: 0.95 μm)	Laser Diffraction	Pass
Wet Sieve Test / 湿筛试验 (75 μm sieve)	$\geq 99.0\%$ through	99.95 %	CIPAC MT 185	Pass
Persistent Foaming / 持久起泡性	$\leq 25\text{ mL}$ after 1 min	10 mL	CIPAC MT 47.2	Pass
Pourability / 倾倒性 (Rinsed Residue)	Residue $\leq 2.5\%$, Rinsed $\leq 0.3\%$	Residue: 1.2 %, Rinsed: 0.12 %	CIPAC MT 148.1	Pass
Heat Stability (54 $\pm$ 2°C, 14 days) / 热储	AI Degradation $\leq 3.0\%$, Suspensibility $\geq 92\%$	Bifen Deg: 0.8%, Etox Deg: 0.5%, Susp: 96.5%	CIPAC MT 46.3	Pass
Low Temp. Stability (0 $\pm$ 2°C, 7 days) / 冷储	No crystal separation or flocculation (无结晶分层)	Conforms / 均相澄清	CIPAC MT 39.3	Pass

CONCLUSION / 检验结论: PASS - Fully complies with FAO Specifications & Enterprise Quality Standard. Approved for release. (合格 - 各项理化指标均符合 FAO 规格及企业内控标准, 准予放行出厂。)

Analyst / 检测员:
Zhang Wei (张伟)
Date: 2026-08-27

Reviewer / 复核员:
Li Ming (李明)
Date: 2026-08-27

QA Director / QA总监:
Dr. Alex Chen (陈博士) [Approved]
Date: 2026-08-27

4. SUPPORTING EVIDENCE 2: BATCH RELEASE & OOS PROTOCOL / 佐证文件二：批放行审批与超标处理规程

Release Gate / 质量关卡	Verification Requirements / 审核项目	Status / 状态	Verified By / 审核人
1. Raw Material Gate (IQC)	Active ingredient purity & isomer ratio, GR carrier attrition resistance, fluorinated container barrier check	Verified Pass	IQC Lead
2. Process Parameters (IPQC)	SC bead mill temp < 32°C, GR drum de-dusting efficiency < 0.2%, CS encapsulation efficiency check	Verified Pass	IPQC Officer
3. Laboratory FQC (COA)	HPLC dual-active assay, laser particle distribution, accelerated thermal/cold stability validation	Verified Pass	Lab Supervisor
4. Packaging & Traceability	2D QR code serialization, induction seal leak-proof testing, occupational vapor limit compliance	Verified Pass	Packaging QA

ACARICIDE / NEMATOCIDE OOS & QUALITY PROTOCOL: Any out-of-specification result triggers automated line lockdown in the ERP. Investigations focus on thermal degradation during bead milling, carrier binder failures in granular coating, or unencapsulated pesticide leakage. Reprocessing or disposal must be jointly authorized by the Factory General Manager and Quality Assurance Director.

杀螨剂与杀线虫剂超标（OOS）调查处置规程：凡任一指标出现偏差，ERP 系统自动锁定出厂闸口并张贴物理红卡。重点排查砂磨温升破坏、颗粒剂包膜黏合剂失效或微囊游离泄漏等根本原因。返工或重组方案必须经工厂总经理与质量总监双重书面签批后方可执行。